UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PRE CALCULUS
COURSE CODE	MAT133
EXAMINATION	OCTOBER 2026
TIME	2 HOURS

NSTRUCTIONS TO CANDIDATES

1. This question paper consists of 5 questions.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) a one – page Appendix 1
   iii) an Answer Booklet - provided by the Faculty
   iv) a graph paper - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 7 printed pages

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QUESTION 1

a) Solve the inequality $5 - 3x \leq -3$. Express the answer in interval notation.

(4 marks)

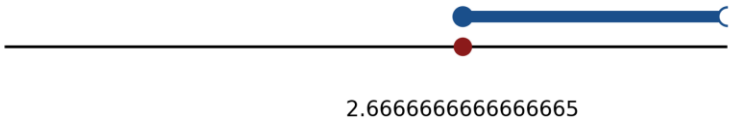


Figure: solution on a number line.

b) Solve the inequality $|3x + 3| > 4$. Express the answer in interval notation and on a number line.

(5 marks)



Figure: solution on a number line.

c) Solve the equation $|5x + 3| = 8$.

(4 marks)

QUESTION 2

a) Apply graph-transformation techniques to sketch $y = -3(x - 4)^2 + 5$. State the vertex and the y-intercept.

(5 marks)

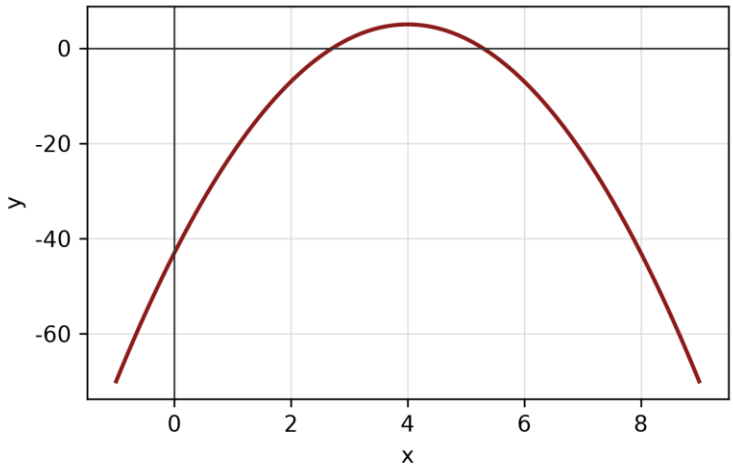


Figure: transformed graph.

b) The piecewise function is defined by
$f(x) = -3x - 3$, if $x < -1$;
$f(x) = 5$, if $-1 \leq x \leq 2$;
$f(x) = x^2 - 3$, if $x > 2$.
Sketch the graph and find f(-2), f(0) and f(3).

(5 marks)



Figure: piecewise graph.

c) The endpoints of a diameter of a circle are (-1, -4) and (4, 5). Find the centre, radius and equation of the circle.

(5 marks)

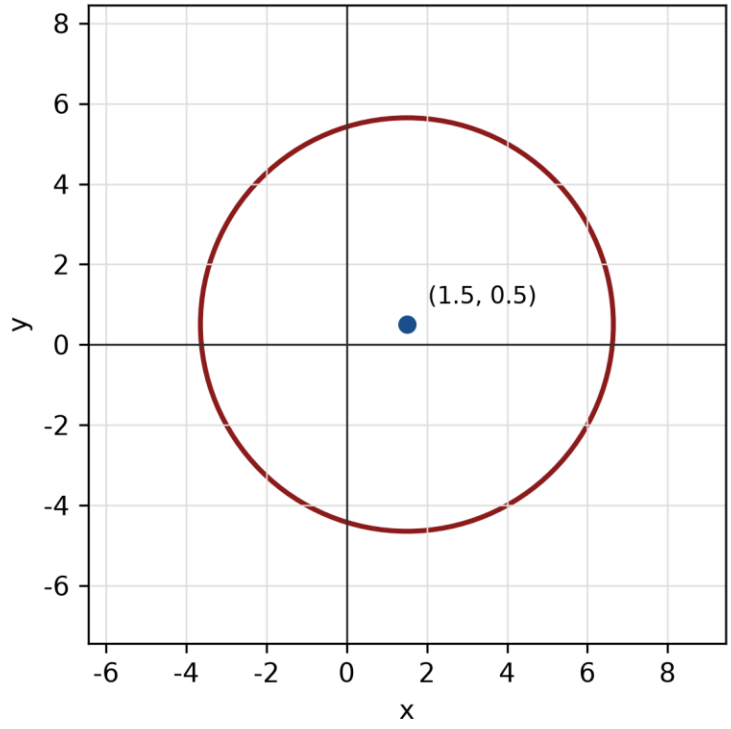


Figure: circle from diameter.

QUESTION 3

a) Given $z_1 = 1 - 4i$ and $z_2 = -2 - i$, express z_1/z_2 in the form a + bi.

(5 marks)

b) Find the modulus and argument of z = -8 + 15i. Give the argument in degrees correct to one

decimal place.

(5 marks)

c) Use long division to find the quotient and remainder when f(x) = -3x^3 + x² - 2x + 4 is divided by x + 4.

(5 marks)

QUESTION 4

a) Solve the system of equations simultaneously:
-2x - 2y - 3z = 0
-x + 2y + 4z = -12
3x - 2y + z = 20

(5 marks)

b) Solve the simultaneous equations
-5x + y = 8
4x - 5y = 2.

(4 marks)

c) Write the equation of the circle with centre (-4, -2) and radius 2. Hence write its general equation and sketch the circle.

(5 marks)

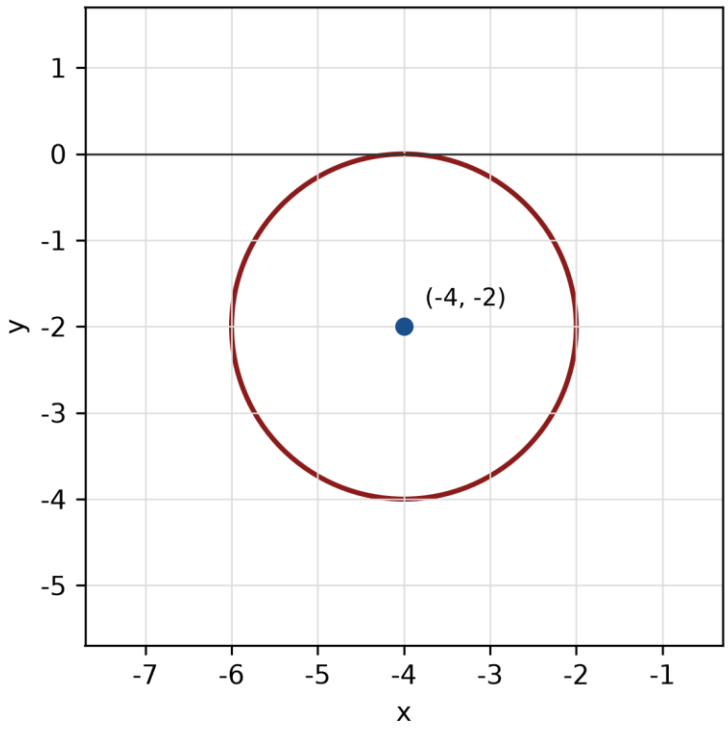


Figure: circle.

QUESTION 5

a) Solve the trigonometric equation -cos² x = 0 for 0° ≤ x < 360°.

(5 marks)

b) Find the area of the sector of a circle with radius 6 cm and central angle 150°.

(4 marks)

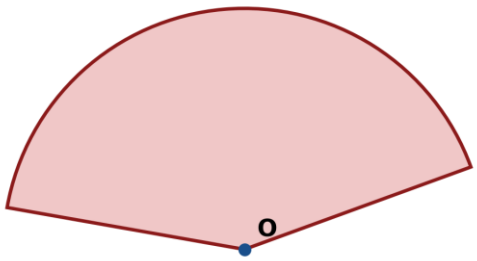


Figure: sector.

c) Solve the triangle with sides a = 5 cm, b = 12 cm and c = 13 cm. Find all three angles.
(5 marks)

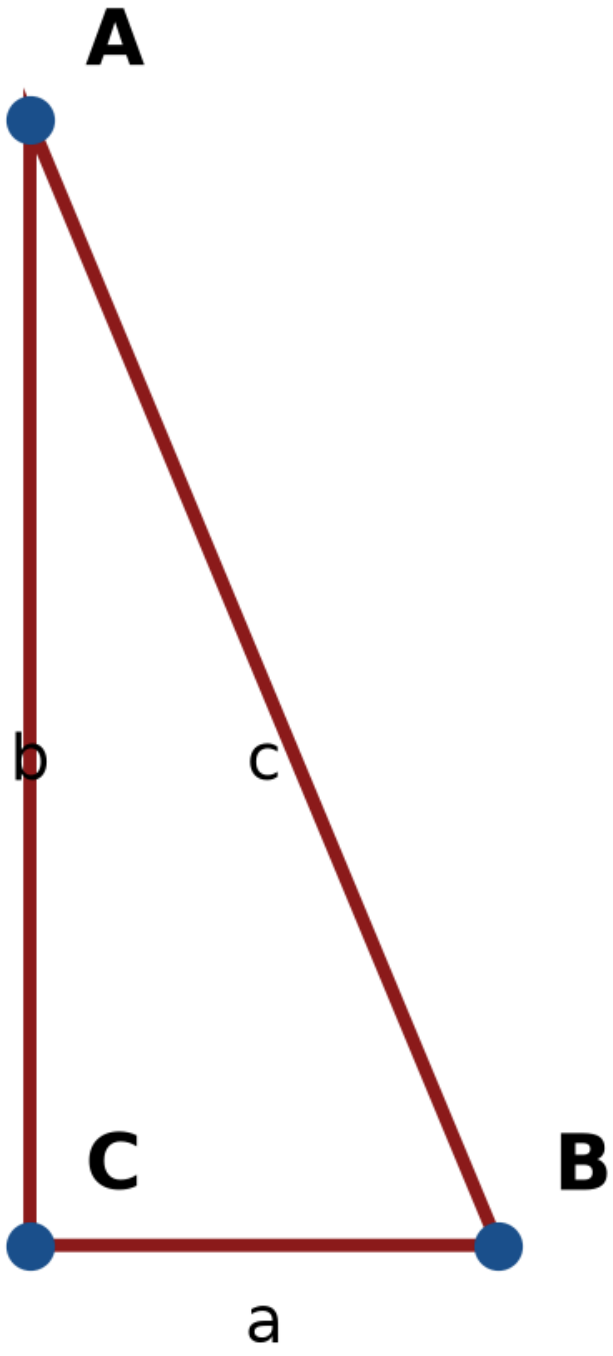


Figure: right triangle.

END OF QUESTION PAPER

APPENDIX 1

Formula sheet (as supplied with the paper).

$\sin^2 A + \cos^2 A = 1$

$\sin(A + B) = \sin A \cos B + \cos A \sin B$

$\cos(A + B) = \cos A \cos B - \sin A \sin B$

$\tan(A - B) = (\tan A - \tan B) / (1 + \tan A \tan B)$

$\cos 2A = \cos^2 A - \sin^2 A$

$\cos 2A = 1 - 2\sin^2 A = 2\cos^2 A - 1$

$\sin 2A = 2 \sin A \cos A$

$\tan^2 A + 1 = \sec^2 A$

$\tan 2A = 2 \tan A / (1 - \tan^2 A)$

Arc length, $s = r\theta$

Area of sector = $\frac{1}{2}r^2\theta$

$a / \sin A = b / \sin B = c / \sin C$

$a^2 = b^2 + c^2 - 2bc \cos A$

Area of triangle = $\frac{1}{2}ab \sin C$

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